Mission-aware cyber incident response generation using reinforcement learning

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***Abstract*—Cybersecurity in military missions is paramount for safeguarding critical assets and ensuring mission success. As cyber threats become increasingly sophisticated, the need for adaptive and optimal defense strategies is essential. This paper introduces a Reinforcement Learning based Markov Decision Process approach designed to model and mitigate cyber attacks within military operations. By leveraging simulations of the the company's critical infrastructure topologyin the Cyber Security Learning Environment , we show that this approach can provide a robust platform for evaluating and optimizing defensive responses against dynamic threat scenarios. The main novelty of this approach being the inclusion of mission-centric risk assessments in the reward function used to train the defence agent.**

The underlying idea is to offer a strategic tool to anticipate, detect, and neutralize cyber adversaries effectively. By simulating realistic attack vectors and adaptive defense mechanisms, this research provides insights and recommendations that can enhance the cybersecurity posture of military alliances, although not yet integrated with actual defense systems. Specifically, the framework can help in real-time threat detection, response optimization, and resilience against sophisticated cyber threats, thereby supporting overarching goals of ensuring collective security and operational integrity.

***Index Terms*—Cybersecurity, Reinforcement Learning, Markov Decision Process, Military Operations, Cyber Defense Strategies**

I. Introduction

Modern military operations are critically dependent on Information and Communication Technology (ICT) systems, which serve as the backbone for mission planning, execution, and decision-making. This reliance introduces significant vulnerabilities as adversaries employ increasingly sophisticated cyber threats aimed at disrupting operations and compromising sensitive data. At the same time, it is crucial that cyber defence strategies align with operational mission objectives while mitigating cyber risks. Not doing so, may result in cyber response policies which can be just as harmful to the mission as the cyber attacks themselves. Shutting off a a server controlling a drone to prevent information leakage, at the cost of losing control of the drone in a densely populated area, might not be the best course of action.

In this paper we introduce a novel RL-based approach that leverages a Markov Decision Process (MDP) [22] and integrates the underlying military mission and its operational objectives in the definition of what an optimal response should be. The work is part of the wider AInception project[1], funded by the European Defence Fund (EDF). AInception explores and develops novel AI-driven tools and techniques for defensive military cyber operations, including detection, analysis and response to adversarial activities. The focus of this paper is on the response phase of the defensive cyber operation.

AInception has taken a scenario-driven approach, where the tools and techniques are developed for particular military missions and storylines. Our work is placed in the following simplified mission called Operation Lighthouse:

The military mission (Operation Lighthouse) The mission has been (post-hoc) developed using an existing dataset [9]. It takes place in a military operation called Operation Lighthouse to a Rapid Deployment Force (RDF) Joint Intelligence Preparation of the Operational Environment (JIPOE) development. An operational liaison and reconnaissance team (OLRT) is deployed locally in a very low-key manner, and can access their ANALYST network over the Internet by VPN. A reach-back element at the RDF Joint HQ is also connected to this ANALYST network. The ANALYST network can be seen in Fig. 2, and is the same network as used to generate the underlying dataset in [9]. It is essentially an ordinary (unclassified) office network, but may at any given point in time contain sensitive data. ANALYST has a central role in supporting military contingency planning activities including meetings where significant decisions will be made, requiring a high level of availability in certain phases. The network also contains information that needs to be kept confidential until certain ministerial processes have run their course. For our purposes, an alert was triggered at a specific time in the reach-back segment of the ANALYST network, after detecting reverse shell

[1] https://www.ainception.eu/

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activity, which we need to find a suitable mission-aware response to.

We use Operation Lighthouse as a case study to illustrate the two main contributions of this paper, which are:

- The extension of a framework for modeling mission-centric ICT risk and security with attributes and metrics, so that it is amendable for reinforcement learning.
- An RL-based approach leveraging MDP that takes into account the provided attributes to generate responses optimized for a given mission.

While our approach is yet not fully integrated into existing military cyber security systems, its modular design and mission-oriented approach hold significant potential for real-world applications. Our results underscore the importance of aligning cyber defense strategies with military operational objectives, ensuring that responses are both context-aware and effective.

The next section provides necessary background and some related work; section III describes the risk framework and our extension of it; section IV describes our RL-based methodology with the implementation described in section V and the results described in section VI; section VII concludes the paper.

II. Background and related work

Automated security response systems, in general, struggle to address the limitations of rule-based approaches to intrusion response. These systems often rely on static rules configured by domain experts, making them labor-intensive, error-prone, and unable to scale effectively to manage the complexity of modern ICT infrastructures. Therefore other approaches like Reinforcement Learning (RL) being used to optimize defensive responses to cyber incidents (see e.g. [10]–[12]). However, we are not aware of RL approaches integrating mission-aware risk in their optimization strategies.

Additionally, data-driven methodologies that integrate digital twins and game-theoretic modeling have also been proposed to solve some related problems. Digital twins provide virtual replicas of target infrastructures, enabling realistic emulation of system behavior, intrusion scenarios, and defensive responses [20]. In military applications, digital twins are extensively used for predictive maintenance of vehicles, aircraft, and satellite systems, optimizing performance, reducing costs, and ensuring operational readiness. They enable real-time simulations for decision-making, realistic training environments, and life-cycle management of military assets, significantly improving efficiency and resilience against evolving cyber and physical threat [23]. By combining these capabilities with game-theoretic optimization, adaptive and efficient intrusion response strategies can be developed and validated.

The *Cyber Security Learning Environment (CSLE)* is an open-source framework designed to address critical challenges in cybersecurity research and implementation. It serves as a high-fidelity and scalable platform for testing, training, and evaluating intrusion detection and response strategies. It aims to bridge the gap between theoretical cybersecurity research and real-world applications by providing realistic environments for developing advanced defense mechanisms.

CSLE consists of two core components: the *emulation system* and the *simulation system*, which together provide a comprehensive testing environment for cybersecurity research. The emulation system in CSLE is designed to recreate real-world IT infrastructures with high fidelity. Using technologies such as Docker containers and virtual switches, it can emulate complex network topologies, including realistic hardware and software configurations. Key capabilities of the emulation system include simulating cyber-attacks such cyber kill chain attacks, generating system logs and metrics to evaluate network performance and response strategies, and providing maintainable datasets that enable researchers to train, validate, and benchmark cybersecurity algorithms.

The simulation system complements the emulation environment by modeling attack-defense scenarios using MDPs. Through MDPs, the simulation system models the states, actions, and transitions of a network under attack, trains RL agents to adapt dynamically to evolving threats, and provides a scalable approach for evaluating the effectiveness of various defensive strategies. The combination of these two systems ensures that CSLE can simulate both theoretical and practical cybersecurity scenarios, enabling users to assess real-world applications. sophisticated cyber threats.

A core feature of CSLE is its use of *Markov Decision Processes (MDPs)* for decision-making under uncertainty. MDPs are mathematical models that provide a structured way to represent cyber scenarios by defining:

- **States:** The conditions of the network, such as "fully operational," "partially compromised," or "under attack."
- **Actions:** Defensive responses that can be taken, such as isolating compromised nodes, blocking traffic, or enhancing monitoring.
- **Transitions:** Probabilities of moving from one state to another after taking an action.
- **Rewards:** Numerical values associated with actions, quantifying their effectiveness in mitigating threats or ensuring operational priorities.

For example, in a scenario where a critical database server is compromised, an MDP might guide the decision to either isolate the server (to prevent further damage) or enhance monitoring (to maintain availability). By balancing short-term and long-term impacts, MDPs allow CSLE to train reinforcement learning agents capable of selecting optimal defensive actions.

When it comes to modeling mission-aware risk in military ICT systems, there have been proposed various frameworks [13], [14], [17]. They all have the same underlying traits, like for instance: the need to model the dependencies between a mission, its tasks, and the technical functions implemented by the ICT; the top-down assessment of the value of the ICT based on the mission goals; and the bottom-up propagation of risk from the ICT to the mission. Most of these approaches, however, are meant to design secure systems rather than supporting cybersecurity during an operation. Some work closer to our needs could be that of Martinez et al. [16], where the goal is

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to produce a real-time assessment of the most mission-critical ICT assets and the threats they are exposed to. This could constitute a good starting point to assess possible response options to cyber incidents taking mission criticalities into account. For our work, instead, we used the mission-centric framework developed at the Norwegian Defence Research Establishment (FFI) [18], which we specifically extended to experiment with RL for cyber response generation in the context of the AInception project.

III. The mission-centric framework

In response to a cyber incident at the ICT level, the course of action should be guided by the operational values that need to be protected and achieved during the military mission. To facilitate this, we introduce a top-down hierarchical model to identify the value chain, establishing traceability between high-level operational objectives (values) and the ICT systems that the military uses to accomplish these objectives.

The top-down model is developed through four analysis steps:

- Modeling of military activities and their objectives
- Modeling of ICT-based functions
- Modeling of ICT system topology
- Modeling of key cyber terrain (value assessment)

This framework was introduced in [17], [18], and in this paper it is extended with a fifth step to explicitly quantify the bottom-up risk and impact-aware assessments:

- Risk and impact-aware evaluation of cyber incidents and response

This extension seeks to address the challenge of quantifying risk in a way suited for the methodology described in the next section. The main idea is to attach attributes to the various levels of the model that can be used to measure and translate the value of an asset at one level to the level below, and likewise express the risk as a reduction of this value. In this way, the value of an operational goal can be associated to the value of a mission task, and in turn to the value of the confidentiality, integrity or availability of an ICT system. We illustrate this through the modeling of the operation Lighthouse which will be also used in the following methodology section.

A. Military mission and objectives

Within Operation Lighthouse, the OLRT is tasked to support the Commander's Decision brief at Friday 13th 0900 AM by providing the appropriate JIPOE products. The OLRT uses the unclassified ANALYST network to collect data and develop the JIPOE products. OLRT accesses the ANALYST network over the Internet by VPN. RDF Joint HQ is connected to the ANALYST network. This is an ordinary office network containing only unclassified data, but the data could be sensitive. Internet can be accessed from the ANALYST network through proxies. In addition to data for developing JIPOE products, the ANALYST network contains OLRT work documents and information which is sensitive and should not leak. For the purpose of our risk assessment, we choose the following two objectives to be of significance: (i) support the Commander's Decision brief at Friday 13th 0900 AM by providing JIPOE products containing correct and available data, and (ii) keep information about cooperation between OLRT and Rotso entities secret. For simplicity reasons, we only focus on the first objective in the following sections

The operational value (V) is given by the ability to deliver these products in time for the Commander´s Decision brief at a given deadline, but also other aspects of the products are important and give operational value. We evaluate the following four operational value attributes:

- *Timeliness* (T) expresses how the timing of mission related information and services contributes to achieving operational objective.
- *Reliability* (R) expresses how the reliability of different kinds of information types and services used by the mission contributes to achieving operational objective.
- *Completeness* (C) expresses how the degree of completion of mission related information and operational tasks contributes to achieving operational objective.
- *Secrecy* (S) expresses how the secrecy of different kinds of information contributes to achieving operational objective.

To quantify operational value through these attributes, we assign them weights representing their relative importance for achieving the operational objectives. The weights of the attributes can assume values from 1 to 4 representing increasing relative importance. The products must be delivered in time or the operation will fail, thus timeliness (T) is the most important attribute for the success of the operation and is assigned the maximum value of 4. Trustworthiness of the information is important since crucial military decision will be based upon these products, thus reliability (R) is given a value of 3, as the second most important attribute. Completeness (C) is considered more important than secrecy (S), although not as critical as T and R. Thus, the weight of C is assigned a value of 2 and the weight of S a value of 1. Hence, the operational value (V) can now be expressed as a combination of the TRCS attributes, weighted by their relative importance in this military operation:

$$V = 4 * T + 3 * R + 2 * C + 1 * S$$

To be able to express the operational value as a number, so that it can be more easily encoded in the reward function of a RL algorithm, we define an easy metric so that each attribute can be assigned one of four values: 0,1,2,3. These numbers represent to which degree the attribute has been fulfilled. Full completion for an attribute would give it a value of 3, so that a report coming in within the deadline will result in T=3. Decision products with a less than 20% uncertainty about the correctness of some of the information they contain, would translate in a reliability score of R=3. If the decision products are based on only half of the available information sources, then it would result in C=1. Similarly, S would be assigned

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a value corresponding to how much of the decision products may have been leaked and be known by the opponent.

Once these scores and metrics are decided, we can quantify the maximum operational value that can be achieved, namely 30, when all attributes have value 3. This allows us also to set a very specific threshold to indicate the risk appetite for a given mission. Only values above this threshold are acceptable, because any lower value would mean that the operational objective was not met. As an example, we could have a threshold of 26, i.e. V should be above 26, which means that if the products are delayed so that T=1, the operational value V will be 26 or less, and consequently below the threshold. For our case, this means that any cyber defence operation should aim at achieving the following to protect the mission:

$$V = 4 * T + 3 * R + 2 * C + 1 * S > 26$$

The four attributes can be used in different ways to describe the operational value. For instance, a threshold can be set for each individual value and a condition that all or some of these thresholds must hold can be used to determine if the operational objectives are achieved. By changing the scores associated with the different attributes, it is possible to prioritize different aspects of the operation in different scenarios. For instance, if secrecy (S) is more important than timeliness (T), their scores could be inverted. Dilemmas may arise, especially when two or more attributes are equally important. If so, then a set of rules to prioritize one over the other(s) in different situations should be defined. For instance, in different phases of an operation, different attributes could be prioritised differently.

B. ICT-based functions

ICT-based functions constitute the use of ICT in military tasks or processes that must be performed to achieve the mission objectives. We identify the following ICT-based functions:

- *Collect:* Collect data from open and closed sources. The compromise of this function will affect the attribute completeness (C). If the collection is interrupted at a certain point, the information basis for the report will be incomplete. If the collection of data fails or delayed it also affects timeliness (T).
- *Send:* Send collected data via the VPN channel from public network to the ANALYST network. The compromise of this function will affect attributes timeliness (T) and completeness (C). Either the data is not transmitted at all, only partially transmitted, or delayed.
- *Analyse and write:* Analyse data against classified needs and requirements and write products. The compromise of this phase will affect the reliability (R) and secrecy (S) if the results of the analysis are changed or leaked. If the analysis and write phase is interrupted, it will affect timeliness (T).
- *Deliver:* Deliver report through secure channels from the intranet to the RDF Joint HQ. The compromise of this phase will affect timeliness (T).

A military operation is often divided in phases or time-periods. The outcome from each phase will determine the degree of success of the overall mission. For simplicity, in this particular case, we assume that all processes are performed sequentially in time, one function used in each phase. This means, for instance, that the Collect process starts at t_0 and ends at t_1, when Send starts. In particular, we assume that if a process is interrupted or compromised at some point, the mission attribute(s) which depends on the process will decrease by one point for each 25% of the process that is left from that point. This type of strict sequential conduct of operational processes will often not be the case in a real situation. The use of ICT-based functions may overlap, be iterative or be connected in other ways, but we consider our assumptions sufficient to develop a proof-of-concept to be used in a reinforcement learning setting. How ICT-based functions affect the various attributes at mission level, and the sequence of the ICT-based functions, are shown schematically in Figure1.

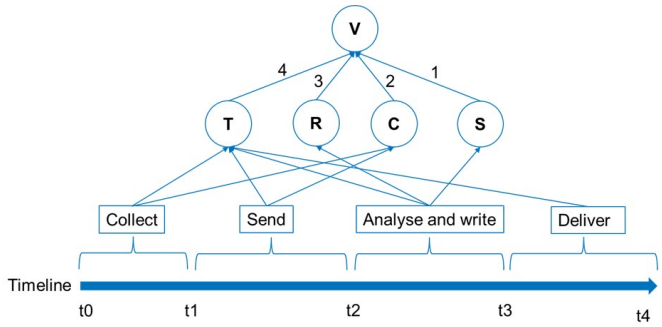


Fig. 1: Schematic description of the mission-centric model with attributes and relationships among them, down to the ICT-based funcions.

C. ICT system topology

The last part of the model is the actual ICT architecture, which needs to be linked to the ICT-based functions so that the security attributes confidentiality (C), integrity (I) and availability (A), can be assigned a value depending on the mission objectives. We do this graphically in Figure 2.

In Table I, we summarize how the ICT-based functions could be compromised, and therefore affect the operational value, based on the compromise of the C, I or A of the ICT systems.

D. Risk and response assessment

It is now possible to define which states of the ICT infrastructure will cause the operational value to drop below that acceptable threshold we defined. In this way it is possible to evaluate both potential threats and response options based on their potential risk for the mission, rather than just for individual ICT systems as schematically illustrated in Figure 3. This, of course, assumes that it is possible to associate a certain detected cyber incident and possible response to how it affects the C, I or A of each system in the ICT infrastructure.

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Time	ICT-based function	C,I,A value
$t_0 \rightarrow t_1$	Collect	Internet and the system used for collection must both be *available (A)* for the OLRT, and the *integrity (I)* preserved, for the whole duration of the Collect phase.
$t_1 \rightarrow t_2$	Send	The link (b) and the VPN server must be *available (A)* and function as intended (*integrity (I)*) during the Send phase.
$t_2 \rightarrow t_3$	Analyze and write	The intranet server must be *available (A)* during the Analyze and write phase, and its *integrity (I) and confidentiality (C)* must not be compromised.
$t_3 \rightarrow t_4$	Deliver	The link from the File Share system to the Cloud server must be *available (A)* during the Deliver phase.

TABLE I: A summary of the relationship between the ICT-based functions, or phases of the operation, and the value of the confidentiality (C), integrity (I) and availability (A) of the ICT systems supporting these phases.

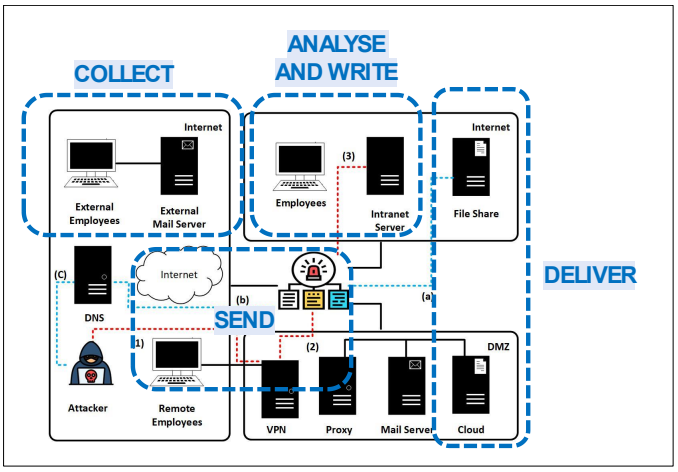


Fig. 2: Digram of the ANALYST network where the ICT-based functions are mapped to concrete systems and network segments. The left most systems are used to collect information from open sources and are connected to the Internet. The systems on the right-up corner are on the intranet and are used to develop the products. The bottom systems are common services in the DMZ among which a cloud server for file sharing.

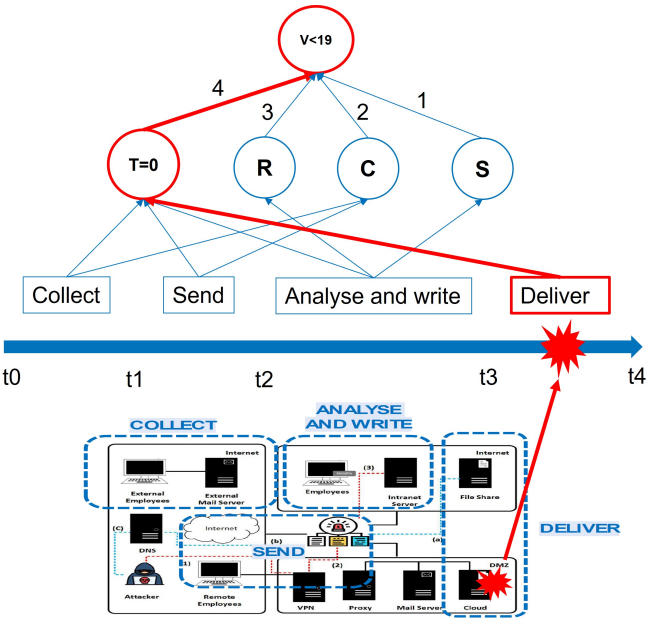


Fig. 3: Example of risk assessment of an incident at time t_3 which prevents the delivery of the products and therefore reduces the value of the timeliness attribute T to 0, so that the overall operational value falls far below the acceptable threshold of 26.

IV. Methodology

Our RL-MDP approach enables dynamic and adaptive responses to evolving cyber threats, balancing operational priorities with effective risk mitigation. A digital twin environment, combining realistic network simulations and modeled attack scenarios, ensures the robustness of the methodology by replicating complex operational conditions. This is combined with the mission-aware risk assessment framework presented in the previous section. We model the cyber defense scenario as a Markov Decision Process (MDP), characterized by:

- **States** (S)**:** Represent different operational phases and detected attack conditions.
- **Actions** (A)**:** Define the set of possible defensive responses.
- **Transition Probabilities** (P)**:** Govern the likelihood of moving from one state to another based on actions taken.
- **Reward Function** ($R(s,a)$)**:** Assigns rewards or penalties based on the effectiveness of actions in mitigating attacks.

The states encapsulate the stages of the cyber operation:

- S_0: Normal Operation
- S_1: Collect Compromise (High Risk) - VPN access alert
- S_2: Send Compromise (Moderate Risk) - Exfiltration alert
- S_3: Analyze & Write Compromise (High Risk) - Internal server compromised
- S_4: Deliver Compromise (Low Risk)
- S_F: Failure State - Mission-critical failures with negative reward
- S_S: Success State - Mission objectives met with positive reward

A. Definition of the Policy π

A policy π is a mapping from states s to probabilities of selecting each action a. Formally, we write:

$$\pi(a \mid s) = \Pr\{A_t = a \mid S_t = s\},$$

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meaning that, when the system is in state s, the policy π prescribes action a with probability $\pi(a \mid s)$.

Our goal is to learn an *optimal policy*, π^*, that maximizes the expected long-term reward. Concretely, we aim to find π^* such that it balances (i) minimizing cyber risk and (ii) preserving mission-critical objectives (e.g., timeliness, secrecy, availability).

$$\pi^* = \arg\max_{\pi} \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t)\Big], \tag{1}$$

where $\gamma \in (0, 1]$ is a discount factor that prioritizes immediate versus future rewards.

For our example, we have restricted the possible defensive actions available to the RL agent to the following:

- A_1: Kill Malicious Process
- A_2: Increase Logging
- A_3: Block Outbound Connections
- A_4: Network Traffic Filtering
- A_5: Take Down ANALYST Network

The reward function $R(s, a)$ is designed to evaluate the RL agent's effectiveness in mitigating cyber threats during military missions. It is mathematically represented as:

$$R(s,a) = MT \times M_T + MR \times M_R + MC \times M_C + MS \times M_S - \beta \times (\text{Risk Penalty}) - \gamma \times (\text{Response Cost}) \tag{2}$$

The mission is partly encoded through the attributes described in section III with the following meaning (and weights for illustration):

- MT (Timeliness) = 4: Determines the importance of responding quickly to detected threats.
- MR (Reliability) = 3: Measures the importance of maintaining data accuracy and integrity during responses.
- MC (Completeness) = 2: Captures the importance of ensuring all necessary data and responses are considered.
- MS (Secrecy) = 1: Reflects the criticality of protecting sensitive information during operations.

The *risk penalty (β)* encodes the impact of compromises. It represents the actual negative impact of the current threat or situation, calculated based on factors such as threat severity, system vulnerability, and potential damage. In missions where avoiding data exfiltration (outbound flow of sensitive information) is highly important, β is set to a high value. This makes the risk penalty more impactful on the overall reward calculation, prioritising actions taken to reduce risk. Conversely, if the mission can tolerate minor data loss or compromise, β is set lower, making the penalty less significant in decision-making.

The *response cost (γ)* represents operational disruptions resulting from defensive actions.

The policy used for action selection is a *softmax function*:

$$P(a|s) = \frac{\exp(Q(s,a)/\tau)}{\sum_{a'} \exp(Q(s,a')/\tau)} \tag{3}$$

The policy is trained to balance exploration of new actions and exploitation of known high-reward actions. Actions are chosen dynamically based on expected rewards, not pre-defined as optimal.

Algorithm 1 RL-Based Optimal Response Action Selection

Require: Current State s, Q-Values $Q(s, a)$, Temperature τ
Ensure: Selected Action a^*

```
Compute softmax probabilities for each action:
for each action a in A do
    P(a|s) ← exp(Q(s, a)/τ)
end for
Normalize probabilities:
P(a|s) ← P(a|s)/ Σ_{a'} exp(Q(s, a')/τ)
Select action based on probabilities:
a* ← Sample(A, P(a|s)) return a*
```

Algorithm 1 shows the pseudocode for the RL-based decision making. The RL agent simulates episodes against evolving attacker strategies to learn optimal actions. *Co-evolutionary learning* enables simultaneous improvement of attacker models and defensive strategies, fostering adaptive decision-making. The policy π is continuously updated based on rewards.

V. Implementation

Our approach to achieving the most effective security response within an ICT infrastructure revolves around the use of a digital twin – a precise virtual representation of the system. By employing this digital counterpart, we simulate various attack scenarios and defensive actions. The outcomes of these simulations yield system metrics and logs, which we then analyze to derive key statistical insights about the infrastructure. These insights enable us to create a mathematical model of the actual system through the process of system identification. We then leverage this model to optimize response strategies, whose performance is assessed using the digital twin. This closed-loop process can be executed iteratively to provide progressively better response strategies.

The topology for the NDT comprising 9 internal employee nodes, 3 remote employee nodes, 6 external user nodes, and 2 mail servers. To implement this in the CSLE, internal employee nodes can be emulated as Docker containers representing workstations or internal endpoints with services like SSH or database access. Remote employee nodes can simulate secure external connections using VPNs or external traffic simulators, while external user nodes can mimic public-facing or untrusted traffic using web or email services with potential vulnerabilities for security testing. Mail servers can be emulated using CSLE's pre-configured SMTP/POP3 Docker images. The santos.com network can be defined as a dedicated virtual topology in CSLE, with virtual OpenFlow switches, traffic generators, and monitoring agents to simulate realistic user behaviors and collect system metrics for analysis. This setup supports testing security policies, intrusion detection, and overall infrastructure resilience

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TABLE II: Examples of Simulated Defender Actions

Index	Action	MITRE D3FEND Technique
1	Kill malicious processes to prevent unauthorized execution.	D3-PS Process Suspension
2	Increase logging to capture suspicious activity for forensic analysis.	D3-FTA Forensic Logging and Analysis
3	Block outbound connections to stop data exfiltration.	D3-ITF Inbound Traffic Filtering
4	Take down network to isolate compromised systems.	D3-NI Network Isolation

TABLE III: Examples of Simualted Attacker Actions

Index	Action	MITRE ATT&CK Technique
1	Perform network scanning to identify live hosts and open ports	Reconnaissance: Active Scanning (T1595.002)
2	Exploit web application vulnerabilities to gain initial access	Initial Access: Exploit Public-Facing Application (T1190)
3	Deploy a webshell for remote command execution	Persistence: Web Shell (T1505.003)
4	Dump password hashes from the compromised system	Credential Access: OS Credential Dumping (T1003)
5	Escalate privileges to administrator or root level	Privilege Escalation: Abuse Elevation Control Mechanism (T1548.002)
6	Exfiltrate sensitive data using DNS tunneling	Exfiltration: Exfiltration Over Alternative Protocol (T1048.002)

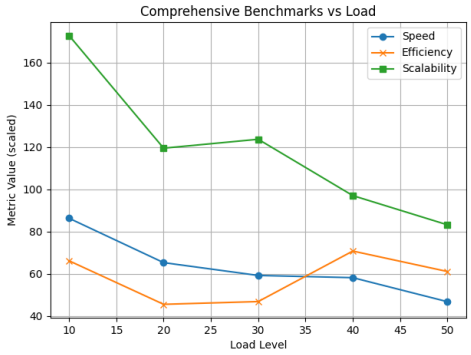


Fig. 4: Comprehensive Benchmarks vs Load.

VI. Results and Discussion

To illustrate our approach, we assume the detection of reverse shell activity originating from `10.229.1.138` (Attacker Node) targeting `192.168.104.1` (Intranet Server) in the *Analyze and Write* phase of a mission. The following response actions are generated:

1) **Terminate Malicious Process:** Immediately halted the reverse shell on the compromised node using the following command:

```
ssh admin@192.168.104.1
       pkill -f reverse_shell
```

2) **Prevent Data Exfiltration:** Applied firewall rules to block outbound traffic to the attacker node:

```
iptables -A OUTPUT -d 10.229.1.138
      -j DROP
```

3) **Notify Security Teams:** Issued alerts to the security team, ensuring stakeholders were informed of the response and its progress:

```
mail -s "Security Alert: Reverse
   Shell Activity" security@domain.com
```

The response actions, dynamically selected by the RL-agent, mitigated the threat while preserving the operational value above the mission's defined threshold ($V > 26$). This ensured the mission objectives were achieved without significant impact on critical timelines or information integrity.

Figure 4 illustrates the comparative performance of speed, efficiency, and scalability as a function of load levels. Scalability sharply declines as load levels increase, indicating system degradation under high stress. This result highlights a critical limitation in the system's ability to handle growing operational demands. In the context of military missions, such performance drop-offs could compromise the timeliness and reliability of data processing during critical operations. For example, during the "Collect" phase of Operation Lighthouse, high load on data collection systems could delay the availability of information needed for analysis, increasing mission risk. Addressing scalability issues is essential to maintaining operational readiness, especially when dynamic changes in load are anticipated in high-stakes environments. This result is derived from simulated benchmarks designed to mimic military-grade loads under realistic conditions.

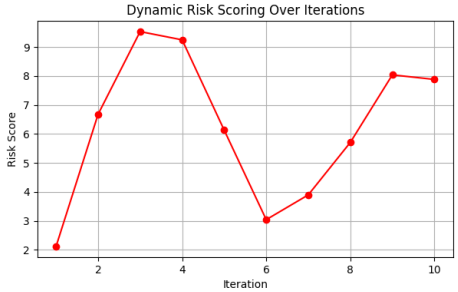


Fig. 5: Dynamic Risk Scoring Over Iterations.

Figure 5 shows the dynamic risk scoring over 10 iterations. The risk score increases sharply during the early iterations as the system begins to detect and respond to evolving threats. The peak at iteration 4 reflects a period of heightened threat exposure, followed by a steady decline as adaptive mitigation strategies take effect. In military missions, such as Operation Lighthouse, this dynamic scoring approach is critical for identifying high-risk phases, such as during the "Analyze and Write" phase where data sensitivity is highest. By prioritizing risk reduction during these phases, the system ensures that mission-critical attributes, including timeliness and secrecy, are preserved. The results demonstrate the effectiveness of iterative reinforcement learning models in adapting to evolving threat landscapes.

Figure 6 highlights the throughput, delay, and security score over multiple iterations. Throughput shows significant fluctuations early on before stabilizing, reflecting the system's optimization efforts under varying conditions. Delay remains consistently low, which is vital for maintaining the timeliness of operations such as the "Deliver" phase in Operation Lighthouse, where reports must reach decision-makers on schedule. The security score exhibits minor variations, showing

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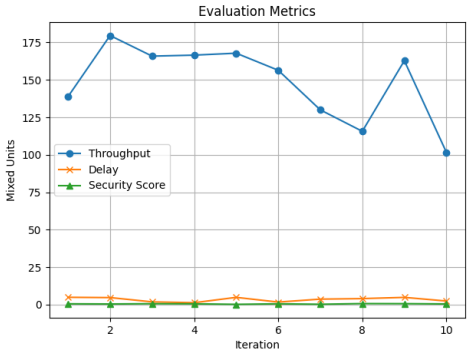


Fig. 6: Evaluation Metrics over Iterations.

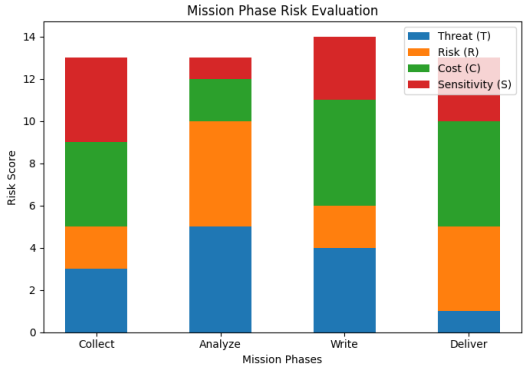


Fig. 8: Mission Phase Risk Evaluation.

resilience to adversarial activities. These results underscore the need to balance throughput and delay while maintaining high security in environments where even minor disruptions could compromise mission success. The metrics are based on real-world simulations of military network performance under varying threat conditions.

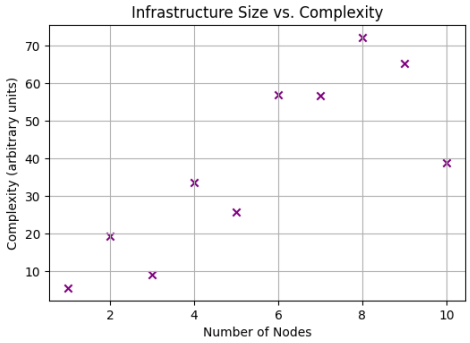


Fig. 7: Infrastructure Size vs Complexity.

Figure 7 correlates infrastructure size (number of nodes) with complexity. The results indicate that complexity increases disproportionately as the number of nodes grows, emphasizing the challenges of managing larger infrastructures. For military missions, this finding is particularly relevant during the ”Send” phase, where multiple interconnected systems must transfer sensitive data securely and reliably. Increased complexity can lead to vulnerabilities, delays, and reduced reliability, which are unacceptable in critical mission contexts. This result highlights the importance of designing scalable and robust architectures to support growing operational demands.

Figure 8 evaluates risk across mission phases. The ”Analyze and Write” phase is identified as the most sensitive and costly, highlighting its criticality in the overall mission. This phase involves handling sensitive data, where any breach or delay could significantly impact the operational value. Threat levels are distributed more evenly across the other phases, but delivery risks remain lower due to the structured nature of data transfer protocols. By encoding phase-specific risks into the reinforcement learning model, the system ensures prioritized allocation of resources and actions to mitigate high-risk scenarios. This approach aligns with the mission requirements of Operation Lighthouse, where achieving timeliness and reliability during data analysis is crucial.

The system achieves a 78.4% success rate, with 21.6% of missions failing due to high operational risk or suboptimal response strategies. This highlights the need for robust risk-aware response mechanisms, especially during high-risk phases like ”Analyze and Write.” The encoded risk assessments in the reward function aim to minimize such failures by dynamically adapting the system’s actions to emerging threats. In military contexts, improving success rates is paramount to ensuring mission objectives are met under strict timelines and operational constraints.

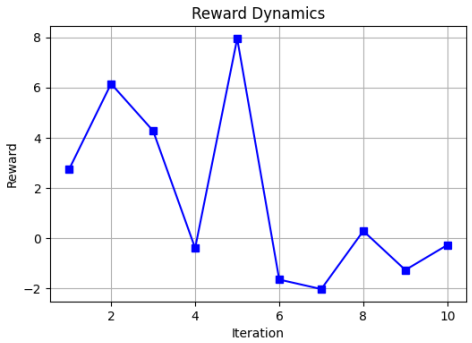


Fig. 9: Reward Dynamics Over Iterations.

Figure 9 depicts reward dynamics across iterations. The fluctuations reflect the system’s learning and adaptation processes, with peaks indicating successful strategy implementations and troughs marking periods of suboptimal decisions. For military missions, such as Operation Lighthouse, these reward dynamics provide valuable feedback to refine response strategies. By aligning rewards with mission-critical attributes like timeliness and secrecy, the system ensures that each iteration moves closer to achieving optimal performance.

Figure 10 shows response times under various scenarios. Adversarial scenarios result in the highest response times, reflecting the challenges posed by sophisticated threats. High-

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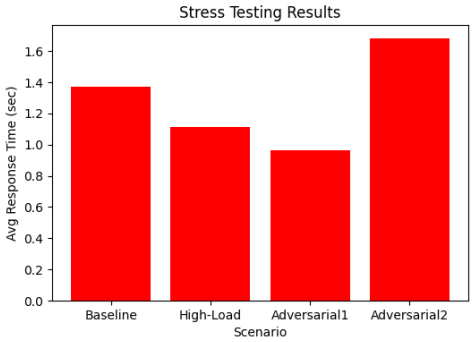


Fig. 10: Stress Testing Results.

load scenarios, while slightly better than baseline conditions, still demonstrate room for optimization. In military missions, such as Operation Lighthouse, these stress testing results highlight the system's ability to maintain functionality under adverse conditions. Ensuring resilience during high-stress periods is crucial for preserving mission integrity and avoiding operational delays.

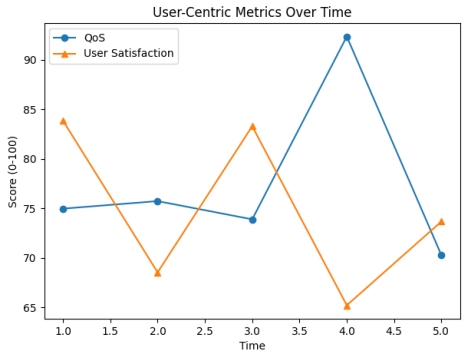


Fig. 11: User-Centric Metrics Over Time.

Figure 11 evaluates quality of service (QoS) and user satisfaction. QoS peaks during optimal phases, while user satisfaction shows a lagged response, reflecting the time required for users to perceive system improvements. In military contexts, maintaining high QoS directly impacts mission-critical operations, while user satisfaction ensures effective collaboration among personnel. Encoding these metrics into the evaluation processes provides a holistic approach to mission success, balancing technical performance with human factors.

Figure 12 illustrates a specific course of action implemented to mitigate data exfiltration risks, leveraging Mitre's D3FEND framework[2]. In this scenario, a Suricata alert flagged a TCPv4 invalid checksum with a risk score of 30 on a primary DNS server running Ubuntu 22.04. To address this, the action `Block Outbound Connections` was executed with the command `iptables -A OUTPUT -j DROP`, effectively preventing outbound communications and mitigating the

[2]https://d3fend.mitre.org/


```
"timestamp": "2025-01-17T20:25:59.515821Z",
"src_ip": "91.189.91.39",
"src_role": "",
"src_os": "",
"src_tools": [],
"dest_ip": "10.229.2.216",
"dest_role": "Primary DNS Server",
"dest_os": "Ubuntu 22.04",
"service": "",
"alert": {
    "action": "blocked",
    "signature": "SURICATA TCPv4 invalid checksum",
    "category": "Generic Protocol Command Decode",
    "risk_score": 30
},
"action_name": "Block Outbound Connections",
"action_description": "Prevent data exfiltration by blocking outbound comms.",
"d3fend_id": "D3-BLOCKOUT",
"action_command": "iptables -A OUTPUT -j DROP",
"response_duration_seconds": 30.0,
"reward": 977.2,
"defender_state": {
    "steps_done": 51,
    "logging_level": 5,
    "agent_type": "DEFENDER",
    "player_type": "ADMIN",
    "abstract_state_number": 4
```


Fig. 12: Course of Action Based on Risk and D3FEND Example.

threat. The response duration was 30 seconds, and the action yielded a reward of 977.2, reflecting its success in maintaining system integrity.

This course of action demonstrates the integration of risk-based decision-making with actionable security measures. Such strategies are critical in military missions, where timely responses to high-risk alerts can prevent cascading failures and preserve mission objectives. By encoding these response strategies into reinforcement learning models, the system ensures adaptability and resilience against evolving cyber threats.

VII. Conclusion

This paper presents a proof-of-concept approach to mission-aware cyber incident response generation using reinforcement learning. The optimal response actions to cyber-attacks against military ICT systems should be guided by the mission's dependence and use of ICT. A mission-centric risk assessment framework has been further developed and used to quantify mission risk and provide input to the reward function used to train the defensive agent. To

By mapping response actions to mission phases, we have demonstrated how simulated environments can be optimized for cyber response that preserve mission-critical attributes, such as timeliness, reliability, completeness, and secrecy in military operations. The RL-based MDP approach showcases its potential to adapt to evolving cyber threats, thereby enhancing the resilience of military mission operations against sophisticated cyber attacks. We have demonstrated our approach for a simplified and fictitious military operation, with promising results. We have yet to address our approach for more complicated and realistic scenarios, as well as address integration with other tools and techniques for defensive cyber operations. This includes integration with, e.g. incident detection and discovery capabilities and analysis capabilities of observed malicious behaviour to develop necessary situational awareness.

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